ASSIGNMENT 7

Sample Data Insertion

Realistic sample data (tables) of the Online Shopping System

1. Objective

This assignment fills every table with realistic data so that the database behaves like a small live store. The data was designed to follow all constraints from Assignment 6 and to contain enough variety for the queries of Assignments 8 and 9: orders in every status, several payment methods and outcomes, delayed and delivered shipments, customers who have never ordered, products that have never been sold, empty and filled carts, and reviews with ratings from 1 to 5.

2. Design of the Data Set

Table	Rows	What the data represents
Pincodes	15	Major Indian cities (Bengaluru, Mumbai, New Delhi, Chennai, Hyderabad, Pune, Kolkata, Ahmedabad) with real PIN-code format
Customers	16	Registered between Jan and Dec 2025; unique e-mails; 10-digit phones; SHA-256 style password hashes (sample values, not real passwords)
Addresses	22	1–2 addresses per customer (Home / Work / Other)
Categories	6	Electronics, Clothing, Footwear, Home & Kitchen, Books, Sports & Fitness
Products	25	Branded products with realistic INR prices and stock levels
Shopping_Cart / Cart_Items	16 / 24	One cart per customer; 10 carts currently hold items, 6 are empty
Orders / Order_Items	28 / 65	17 Delivered, 4 Shipped, 4 Placed, 3 Cancelled; totals equal Σ(quantity × unit_price)
Payments	28	UPI, Card, Net Banking, Cash on Delivery; Successful, Pending and Failed outcomes
Shipments	21	Created only for Shipped and Delivered orders; couriers Delhivery, Blue Dart, Ecom Express, India Post, XpressBees
Reviews	25	Ratings 1–5, written only by customers whose order was Delivered; one review per customer per product

3. Insert Order and Referential Rules

Parent tables are loaded before the tables that reference them: **Pincodes → Customers → Addresses → Categories → Products → Shopping_Cart → Cart_Items → Orders → Order_Items → Payments → Shipments → Reviews**. Loading in any other order would be rejected by the foreign keys. Primary-key values are given explicitly so that foreign-key references in the script are easy to read and verify.

4. INSERT Statements

Header

```
-- ASSIGNMENT 7: SAMPLE DATA INSERTION
USE online_shopping;
```

1. Pincodes (lookup)

```
-- 1. Pincodes (lookup)
INSERT INTO Pincodes (pincode, city, state) VALUES
  ('560001', 'Bengaluru', 'Karnataka'),
  ('560034', 'Bengaluru', 'Karnataka'),
  ('560066', 'Bengaluru', 'Karnataka'),
  ('400001', 'Mumbai', 'Maharashtra'),
  ('400050', 'Mumbai', 'Maharashtra'),
  ('110001', 'New Delhi', 'Delhi'),
  ('110016', 'New Delhi', 'Delhi'),
  ('600001', 'Chennai', 'Tamil Nadu'),
  ('600040', 'Chennai', 'Tamil Nadu'),
  ('500001', 'Hyderabad', 'Telangana'),
  ('500081', 'Hyderabad', 'Telangana'),
  ('411001', 'Pune', 'Maharashtra'),
  ('411045', 'Pune', 'Maharashtra'),
  ('700001', 'Kolkata', 'West Bengal'),
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  ('380001', 'Ahmedabad', 'Gujarat');
```

2. Customers

```
-- 2. Customers
INSERT INTO Customers (customer_id, first_name, last_name, email, phone, password_hash, registration_date) VALUES
  (1, 'Aarav', 'Sharma', 'aarav.sharma@gmail.com', '9977446522', 'f9a770c5c6f811a25626ad6a222734aeb46bfa684ab5ce83eccc3180d4580cae',
'2025-01-14'),
  (2, 'Priya', 'Nair', 'priya.nair@yahoo.in', '6323775684', '7097e38d11e796eb6bd10b4fe71183e136e5ae9e871af3f3c8cb6e932b0cc140',
'2025-02-03'),
  (3, 'Rohan', 'Mehta', 'rohan.mehta@gmail.com', '7093414184', '7a89b7d96d722e3946e212a126755717fde3c9734d61fd4db97aea19580231ae',
'2025-02-21'),
  (4, 'Ananya', 'Iyer', 'ananya.iyer@yahoo.in', '8886848884', '2ae16de3069869902a60caeb23f7693c301b768e284edb203ada77e4344cd48b',
'2025-03-09'),
  (5, 'Vikram', 'Singh', 'vikram.singh@yahoo.in', '6375888073', 'a0cb171bfd8198114d6a63a91040f92bcf77e72a35fa9119ddf84c32c7adbf9d',
'2025-04-17'),
  (6, 'Sneha', 'Reddy', 'sneha.reddy@yahoo.in', '8842203407', '26a7bae8b257597cb091dc116bf424d42ed1c296877c57017480e4fc53f91d32',
'2025-05-02'),
  (7, 'Karthik', 'Rao', 'karthik.rao@gmail.com', '7020834547', 'dea97d04b1b622acde421d1c8962b474832b9841844706ea483677c3874b1de1',
'2025-05-28'),
  (8, 'Meera', 'Joshi', 'meera.joshi@gmail.com', '9923160265', 'f8e8de366a0533fec33fdc3ac561d61344fe6339cebec73da05ea918b81c9b0b',
'2025-06-11'),
  (9, 'Arjun', 'Patel', 'arjun.patel@outlook.com', '9875805171', '68381a62901a5f2c069b9f1b1792103e7068e9cfe236fee13e9dd34f20170676',
'2025-07-06'),
  (10, 'Divya', 'Menon', 'divya.menon@yahoo.in', '9938215876', '19e2910942187180294a8334ac9edc438fccd8e86ccaca4fd1270b6d8e6826d0',
'2025-07-30'),
  (11, 'Imran', 'Khan', 'imran.khan@outlook.com', '9956661846', '92c1eac125f4737114f97a60cb3cd58903021067f8e08830fdb6ac261d195e1b',
'2025-08-19'),
  (12, 'Pooja', 'Bansal', 'pooja.bansal@outlook.com', '8878932553',
'f33be4149a823c0291635e6098630afec9017f5c484d311285ba93a81d81d543', '2025-09-08'),
  (13, 'Siddharth', 'Das', 'siddharth.das@yahoo.in', '9855686986',
'2c1432365b2c92840d7cd1207c6bfcc9fe84e8c1728708f5d42d274ffb91777b', '2025-09-25'),
  (14, 'Neha', 'Kulkarni', 'neha.kulkarni@gmail.com', '7082217824',
'e64f786c8b165fb3a3c1ecf6a447719a9b8ece963d8b286800fa471541965547', '2025-10-13'),
  (15, 'Rahul', 'Verma', 'rahul.verma@outlook.com', '9971397604', 'b898d669104f5dc36da3e97c691302356ecaab50116a42f7fdce31aad6a5ff30',
'2025-11-04'),
  (16, 'Lakshmi', 'Narayanan', 'lakshmi.narayanan@gmail.com', '6387500451',
'095d13832e5795329be92fe9a9f2b2718a80629f1f6a07b41a65292b1ad52ac3', '2025-12-01');
```

3. Addresses

```
-- 3. Addresses
INSERT INTO Addresses (address_id, customer_id, address_line, pincode, address_type) VALUES
  (1, 1, '12, 4th Cross, Indiranagar', '560034', 'Home'),
  (2, 1, 'Tower B, Embassy Tech Village, Outer Ring Road', '560066', 'Work'),
  (3, 2, '45, Residency Road, Near Brigade Towers', '560001', 'Home'),
  (4, 3, 'Flat 302, Sea View Apartments, Colaba', '400001', 'Home'),
  (5, 3, 'Nirmal Tower, Nariman Point', '400050', 'Work'),
  (6, 4, '7, Anna Salai, Near Spencer Plaza', '600001', 'Home'),
  (7, 5, 'B-14, Connaught Place', '110001', 'Home'),
  (8, 5, '22, Hauz Khas Village', '110016', 'Other'),
  (9, 6, 'Plot 56, Banjara Hills Road No. 3', '500001', 'Home'),
  (10, 7, '88, MG Road, Near Trinity Metro', '560001', 'Home'),
  (11, 7, 'Prestige Tech Park, Marathahalli', '560066', 'Work'),
  (12, 8, '21, Koregaon Park Lane 5', '411001', 'Home'),
  (13, 9, 'C-9, Navrangpura Society', '380001', 'Home'),
  (14, 10, '15, Jayanagar 4th Block', '560034', 'Home'),
  (15, 11, 'Flat 8A, Bandra West', '400050', 'Home'),
  (16, 11, 'Nariman Point Business Centre', '400050', 'Work'),
  (17, 12, 'D-33, Hauz Khas Enclave', '110016', 'Home'),
  (18, 13, '5, Park Street', '700001', 'Home'),
  (19, 14, 'Lane 7, Viman Nagar', '411045', 'Home'),
  (20, 14, 'Hinjewadi Phase 1 IT Park', '411045', 'Work'),
  (21, 15, 'Gachibowli Main Road, Flat 12', '500081', 'Home'),
  (22, 16, '31, T Nagar, Near Pondy Bazaar', '600040', 'Home');
```

4. Categories

```
-- 4. Categories
INSERT INTO Categories (category_id, category_name, description) VALUES
  (1, 'Electronics', 'Mobiles, audio, wearables and accessories'),
  (2, 'Clothing', 'Men''s and women''s apparel'),
  (3, 'Footwear', 'Sports, casual and formal footwear'),
  (4, 'Home & Kitchen', 'Appliances, cookware and storage'),
  (5, 'Books', 'Academic, self-help and general reading'),
  (6, 'Sports & Fitness', 'Equipment and fitness accessories');
```

5. Products

```
-- 5. Products
INSERT INTO Products (product_id, category_id, product_name, description, price, stock_quantity, brand) VALUES
  (1, 1, 'Redmi Note 13 5G (8GB/128GB)', '6.67-inch AMOLED display, 5000 mAh battery', 17999, 45, 'Xiaomi'),
  (2, 1, 'Rockerz 450 Bluetooth Headphones', 'On-ear wireless headphones, 15-hour playback', 1499, 120, 'boAt'),
  (3, 1, '25W Fast Charger', 'USB-C wall charger with PD support', 1199, 200, 'Samsung'),
  (4, 1, 'M331 Silent Plus Wireless Mouse', 'Silent-click wireless mouse, 24-month battery', 899, 85, 'Logitech'),
  (5, 1, 'ColorFit Pulse 3 Smartwatch', '1.96-inch display, Bluetooth calling', 1999, 70, 'Noise'),
  (6, 2, '511 Slim Fit Jeans', 'Stretch denim, mid-rise slim fit', 2799, 60, 'Levi''s'),
  (7, 2, 'Men''s Formal Shirt', 'Cotton blend regular-fit full-sleeve shirt', 1499, 90, 'Allen Solly'),
  (8, 2, 'Round Neck Cotton T-Shirt', 'Pure cotton everyday t-shirt', 799, 150, 'Puma'),
  (9, 2, 'Women''s Cotton Kurta', 'Hand-block printed straight kurta', 1899, 55, 'Fabindia'),
  (10, 3, 'Revolution 7 Running Shoes', 'Lightweight cushioned road-running shoes', 3995, 40, 'Nike'),
  (11, 3, 'Leather Formal Shoes', 'Genuine leather lace-up oxfords', 2299, 35, 'Bata'),
  (12, 3, 'Adilette Comfort Slides', 'Contoured footbed slides', 1799, 75, 'Adidas'),
  (13, 3, 'Casual Canvas Sneakers', 'Everyday lace-up sneakers', 1499, 95, 'Campus'),
  (14, 4, 'Induction Cooktop 2000W', 'Push-button induction cooktop', 2499, 50, 'Prestige'),
  (15, 4, 'Thermosteel Flask 1L', '24-hour hot and cold insulation', 799, 110, 'Milton'),
  (16, 4, 'Glass Tumbler Set (6 pcs)', 'Borosilicate 300 ml tumblers', 699, 80, 'Borosil'),
  (17, 4, 'Digital Air Fryer 4.1L', 'Rapid air technology, 8 preset menus', 7999, 30, 'Philips'),
  (18, 5, 'Atomic Habits', 'Self-help paperback by James Clear', 499, 140, 'Penguin Random House'),
  (19, 5, 'The Psychology of Money', 'Paperback by Morgan Housel', 349, 160, 'Jaico'),
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  (20, 5, 'Database System Concepts', 'Silberschatz, Korth and Sudarshan, 7th edition', 899, 60, 'McGraw Hill'),
  (21, 5, 'Wings of Fire', 'Autobiography of A.P.J. Abdul Kalam', 299, 180, 'Universities Press'),
  (22, 6, 'Mavis 350 Nylon Shuttlecocks (Pack of 6)', 'Medium-speed nylon shuttlecocks', 649, 100, 'Yonex'),
  (23, 6, 'Football Size 5', 'Machine-stitched match football', 599, 65, 'Cosco'),
  (24, 6, 'Yoga Mat 6mm', 'Anti-skid EVA yoga mat', 549, 130, 'Boldfit'),
  (25, 6, 'PVC Dumbbell Set 10 kg', 'Adjustable dumbbell set with rods', 1299, 45, 'Nivia');
```

6. Shopping_Cart

```
-- 6. Shopping_Cart
INSERT INTO Shopping_Cart (cart_id, customer_id, created_date) VALUES
  (1, 1, '2025-01-25'),
  (2, 2, '2025-02-19'),
  (3, 3, '2025-02-25'),
  (4, 4, '2025-04-06'),
  (5, 5, '2025-05-14'),
  (6, 6, '2025-05-26'),
  (7, 7, '2025-05-31'),
  (8, 8, '2025-06-24'),
  (9, 9, '2025-07-15'),
  (10, 10, '2025-08-19'),
  (11, 11, '2025-08-29'),
  (12, 12, '2025-09-29'),
  (13, 13, '2025-09-29'),
  (14, 14, '2025-10-25'),
  (15, 15, '2025-11-20'),
  (16, 16, '2025-12-31');
```

7. Cart_Items

```
-- 7. Cart_Items
INSERT INTO Cart_Items (cart_item_id, cart_id, product_id, quantity) VALUES
  (1, 1, 4, 1),
  (2, 1, 18, 1),
  (3, 1, 17, 2),
  (4, 2, 19, 1),
  (5, 2, 12, 1),
  (6, 2, 3, 1),
  (7, 3, 9, 2),
  (8, 3, 25, 1),
  (9, 5, 11, 2),
  (10, 5, 14, 2),
  (11, 6, 11, 2),
  (12, 6, 24, 1),
  (13, 8, 11, 1),
  (14, 8, 6, 1),
  (15, 8, 7, 2),
  (16, 10, 25, 1),
  (17, 10, 12, 2),
  (18, 12, 7, 2),
  (19, 12, 5, 2),
  (20, 12, 15, 2),
  (21, 13, 14, 1),
  (22, 13, 9, 1),
  (23, 14, 9, 1),
  (24, 14, 15, 1);
```

8. Orders

```
-- 8. Orders
INSERT INTO Orders (order_id, customer_id, address_id, order_date, order_status, total_amount) VALUES
  (1, 10, 14, '2026-01-17 10:45:00', 'Delivered', 24379),
  (2, 10, 14, '2026-02-20 18:56:00', 'Delivered', 4591),
  (3, 10, 14, '2026-03-05 20:06:00', 'Delivered', 2590),
  (4, 3, 5, '2026-03-12 21:39:00', 'Delivered', 3238),
  (5, 12, 17, '2026-04-29 16:30:00', 'Cancelled', 3768),
  (6, 1, 2, '2026-05-17 12:30:00', 'Delivered', 13844),
  (7, 13, 18, '2026-06-12 13:54:00', 'Delivered', 1499),
  (8, 3, 5, '2026-07-10 11:40:00', 'Delivered', 6991),
  (9, 2, 3, '2026-07-12 21:29:00', 'Delivered', 19439),
  (10, 14, 20, '2026-07-27 20:30:00', 'Delivered', 38878),
  (11, 7, 11, '2026-07-28 21:59:00', 'Delivered', 4047),
  (12, 8, 12, '2026-08-08 18:16:00', 'Cancelled', 24096),
  (13, 14, 19, '2026-08-09 18:25:00', 'Delivered', 1499),
  (14, 9, 13, '2026-08-10 20:19:00', 'Delivered', 2597),
  (15, 14, 19, '2026-08-29 18:51:00', 'Delivered', 6092),
  (16, 7, 10, '2026-09-07 20:32:00', 'Delivered', 4196),
  (17, 9, 13, '2026-09-13 21:30:00', 'Delivered', 4794),
  (18, 4, 6, '2026-09-17 16:27:00', 'Cancelled', 1647),
  (19, 14, 20, '2026-09-21 13:18:00', 'Delivered', 6142),
  (20, 3, 5, '2026-09-24 14:00:00', 'Delivered', 4197),
  (21, 12, 17, '2026-10-08 08:09:00', 'Shipped', 7491),
  (22, 3, 4, '2026-10-09 18:56:00', 'Shipped', 22297),
  (23, 5, 7, '2026-10-10 17:29:00', 'Shipped', 1499),
  (24, 8, 12, '2026-10-12 20:33:00', 'Shipped', 1499),
  (25, 6, 9, '2026-10-14 15:23:00', 'Placed', 2499),
  (26, 7, 11, '2026-10-18 11:16:00', 'Placed', 22396),
  (27, 13, 18, '2026-10-19 19:31:00', 'Placed', 5994),
  (28, 7, 10, '2026-10-20 15:32:00', 'Placed', 5194);
```

9. Order_Items

```
-- 9. Order_Items
INSERT INTO Order_Items (order_item_id, order_id, product_id, quantity, unit_price) VALUES
  (1, 1, 13, 2, 1499),
  (2, 1, 1, 1, 19439),
  (3, 1, 4, 2, 971),
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  (4, 2, 23, 3, 599),
  (5, 2, 3, 1, 1295),
  (6, 2, 7, 1, 1499),
  (7, 3, 3, 2, 1295),
  (8, 4, 2, 2, 1619),
  (9, 5, 18, 2, 499),
  (10, 5, 4, 1, 971),
  (11, 5, 12, 1, 1799),
  (12, 6, 11, 1, 2299),
  (13, 6, 17, 1, 7999),
  (14, 6, 24, 3, 549),
  (15, 6, 9, 1, 1899),
  (16, 7, 7, 1, 1499),
  (17, 8, 7, 1, 1499),
  (18, 8, 18, 3, 499),
  (19, 8, 10, 1, 3995),
  (20, 9, 1, 1, 19439),
  (21, 10, 1, 2, 19439),
  (22, 11, 5, 1, 1999),
  (23, 11, 2, 1, 1499),
  (24, 11, 24, 1, 549),
  (25, 12, 12, 2, 1799),
  (26, 12, 14, 1, 2499),
  (27, 12, 1, 1, 17999),
  (28, 13, 13, 1, 1499),
  (29, 14, 15, 1, 799),
  (30, 14, 21, 1, 299),
  (31, 14, 2, 1, 1499),
  (32, 15, 4, 2, 899),
  (33, 15, 10, 1, 3995),
  (34, 15, 21, 1, 299),
  (35, 16, 8, 2, 799),
  (36, 16, 25, 2, 1299),
  (37, 17, 10, 1, 3995),
  (38, 17, 8, 1, 799),
  (39, 18, 24, 3, 549),
  (40, 19, 10, 1, 3995),
  (41, 19, 19, 1, 349),
  (42, 19, 20, 2, 899),
  (43, 20, 14, 1, 2499),
  (44, 20, 18, 1, 499),
  (45, 20, 3, 1, 1199),
  (46, 21, 10, 1, 3995),
  (47, 21, 2, 1, 1499),
  (48, 21, 4, 1, 899),
  (49, 21, 24, 2, 549),
  (50, 22, 5, 1, 1999),
  (51, 22, 11, 1, 2299),
  (52, 22, 1, 1, 17999),
  (53, 23, 7, 1, 1499),
  (54, 24, 2, 1, 1499),
  (55, 25, 14, 1, 2499),
  (56, 26, 12, 2, 1799),
  (57, 26, 15, 1, 799),
  (58, 26, 1, 1, 17999),
  (59, 27, 25, 2, 1299),
  (60, 27, 21, 3, 299),
  (61, 27, 14, 1, 2499),
  (62, 28, 15, 1, 799),
  (63, 28, 21, 2, 299),
  (64, 28, 4, 2, 899),
  (65, 28, 5, 1, 1999);
```

10. Payments

```
-- 10. Payments
INSERT INTO Payments (payment_id, order_id, payment_method, payment_status, payment_date, amount_paid) VALUES
  (1, 1, 'Card', 'Successful', '2026-01-17 10:46:00', 24379),
  (2, 2, 'Net Banking', 'Successful', '2026-02-20 18:58:00', 4591),
  (3, 3, 'UPI', 'Successful', '2026-03-05 20:12:00', 2590),
  (4, 4, 'Card', 'Successful', '2026-03-12 21:42:00', 3238),
  (5, 5, 'Net Banking', 'Failed', '2026-04-29 16:32:00', 0),
  (6, 6, 'Net Banking', 'Successful', '2026-05-17 12:33:00', 13844),
  (7, 7, 'Cash on Delivery', 'Successful', '2026-06-16 18:30:00', 1499),
  (8, 8, 'UPI', 'Successful', '2026-07-10 11:46:00', 6991),
  (9, 9, 'Cash on Delivery', 'Successful', '2026-07-18 18:30:00', 19439),
  (10, 10, 'UPI', 'Successful', '2026-07-27 20:31:00', 38878),
  (11, 11, 'Card', 'Successful', '2026-07-28 22:02:00', 4047),
  (12, 12, 'UPI', 'Failed', '2026-08-08 18:18:00', 0),
  (13, 13, 'UPI', 'Successful', '2026-08-09 18:33:00', 1499),
  (14, 14, 'UPI', 'Successful', '2026-08-10 20:26:00', 2597),
  (15, 15, 'Net Banking', 'Successful', '2026-08-29 19:00:00', 6092),
  (16, 16, 'UPI', 'Successful', '2026-09-07 20:41:00', 4196),
  (17, 17, 'Cash on Delivery', 'Successful', '2026-09-19 18:30:00', 4794),
  (18, 18, 'Card', 'Failed', '2026-09-17 16:31:00', 0),
  (19, 19, 'UPI', 'Successful', '2026-09-21 13:27:00', 6142),
  (20, 20, 'Net Banking', 'Successful', '2026-09-24 14:06:00', 4197),
  (21, 21, 'Card', 'Successful', '2026-10-08 08:13:00', 7491),
  (22, 22, 'UPI', 'Successful', '2026-10-09 19:03:00', 22297),
  (23, 23, 'Net Banking', 'Successful', '2026-10-10 17:38:00', 1499),
  (24, 24, 'UPI', 'Successful', '2026-10-12 20:36:00', 1499),
  (25, 25, 'Cash on Delivery', 'Pending', NULL, 0),
  (26, 26, 'UPI', 'Successful', '2026-10-18 11:19:00', 22396),
  (27, 27, 'Cash on Delivery', 'Pending', NULL, 0),
  (28, 28, 'UPI', 'Successful', '2026-10-20 15:34:00', 5194);
```

11. Shipments

```
-- 11. Shipments
INSERT INTO Shipments (shipment_id, order_id, courier_name, tracking_number, shipped_date, expected_delivery_date, shipment_status)
VALUES
  (1, 1, 'XpressBees', 'XB3408887563', '2026-01-18', '2026-01-21', 'Delivered'),
  (2, 2, 'Blue Dart', 'BD1508021414', '2026-02-22', '2026-02-25', 'Delivered'),
  (3, 3, 'XpressBees', 'XB2298115953', '2026-03-07', '2026-03-12', 'Delivered'),
  (4, 4, 'Ecom Express', 'ECM2619985073', '2026-03-13', '2026-03-17', 'Delivered'),
  (5, 6, 'XpressBees', 'XB7678236685', '2026-05-18', '2026-05-23', 'Delivered'),
  (6, 7, 'Blue Dart', 'BD7351583601', '2026-06-13', '2026-06-16', 'Delivered'),
  (7, 8, 'Blue Dart', 'BD9753181212', '2026-07-12', '2026-07-17', 'Delivered'),
  (8, 9, 'Ecom Express', 'ECM8486523758', '2026-07-13', '2026-07-18', 'Delivered'),
  (9, 10, 'Blue Dart', 'BD5166949866', '2026-07-28', '2026-07-31', 'Delivered'),
  (10, 11, 'India Post', 'IP8855719729', '2026-07-30', '2026-08-03', 'Delivered'),
  (11, 13, 'India Post', 'IP2340861552', '2026-08-11', '2026-08-15', 'Delivered'),
  (12, 14, 'Blue Dart', 'BD9377086414', '2026-08-11', '2026-08-14', 'Delivered'),
  (13, 15, 'India Post', 'IP1741194397', '2026-08-30', '2026-09-04', 'Delivered'),
  (14, 16, 'Ecom Express', 'ECM4176214552', '2026-09-08', '2026-09-13', 'Delivered'),
  (15, 17, 'XpressBees', 'XB7796305882', '2026-09-14', '2026-09-19', 'Delivered'),
  (16, 19, 'Blue Dart', 'BD1367377426', '2026-09-23', '2026-09-26', 'Delivered'),
  (17, 20, 'XpressBees', 'XB9768556240', '2026-09-25', '2026-09-29', 'Delivered'),
  (18, 21, 'Blue Dart', 'BD7619254649', '2026-10-09', '2026-10-13', 'Delayed'),
  (19, 22, 'Blue Dart', 'BD7110247693', '2026-10-11', '2026-10-16', 'In Transit'),
  (20, 23, 'Blue Dart', 'BD1126756425', '2026-10-12', '2026-10-17', 'In Transit'),
  (21, 24, 'Blue Dart', 'BD2608711742', '2026-10-13', '2026-10-16', 'In Transit');
```

12. Reviews

```
-- 12. Reviews
INSERT INTO Reviews (review_id, product_id, customer_id, rating, comment, review_date) VALUES
  (1, 13, 10, 5, 'Excellent quality, totally worth the price.', '2026-01-29'),
  (2, 1, 10, 5, 'Superb product and quick delivery!', '2026-01-28'),
  (3, 4, 10, 4, 'Solid build and good performance.', '2026-01-24'),
  (4, 7, 10, 4, 'Solid build and good performance.', '2026-03-06'),
  (5, 17, 1, 5, 'Superb product and quick delivery!', '2026-05-29'),
  (6, 24, 1, 2, 'Quality could be better.', '2026-06-04'),
  (7, 9, 1, 4, 'Good value for money, minor packaging issue.', '2026-05-28'),
  (8, 7, 13, 4, 'Works well. Would recommend.', '2026-06-25'),
  (9, 18, 3, 5, 'Excellent quality, totally worth the price.', '2026-07-29'),
  (10, 10, 3, 4, 'Works well. Would recommend.', '2026-07-21'),
  (11, 1, 2, 5, 'Superb product and quick delivery!', '2026-07-26'),
  (12, 1, 14, 4, 'Solid build and good performance.', '2026-08-04'),
  (13, 2, 7, 5, 'Excellent quality, totally worth the price.', '2026-08-05'),
  (14, 24, 7, 5, 'Excellent quality, totally worth the price.', '2026-08-11'),
  (15, 13, 14, 5, 'Excellent quality, totally worth the price.', '2026-08-20'),
  (16, 15, 9, 3, 'Decent but nothing special.', '2026-08-27'),
  (17, 21, 9, 5, 'Exactly as described. Very happy with the purchase.', '2026-08-21'),
  (18, 2, 9, 4, 'Good value for money, minor packaging issue.', '2026-08-25'),
  (19, 8, 7, 5, 'Exactly as described. Very happy with the purchase.', '2026-09-23'),
  (20, 25, 7, 2, 'Not as durable as expected.', '2026-09-27'),
  (21, 10, 9, 4, 'Works well. Would recommend.', '2026-09-20'),
  (22, 10, 14, 3, 'Okay for the price, expected slightly better.', '2026-10-01'),
  (23, 19, 14, 3, 'Okay for the price, expected slightly better.', '2026-10-04'),
  (24, 20, 14, 5, 'Excellent quality, totally worth the price.', '2026-10-04'),
  (25, 3, 3, 4, 'Works well. Would recommend.', '2026-10-14');
```

5. Execution and Verification

5.1 Script executed

SQL query

```
SELECT 'all inserts succeeded' AS result;
```

Output

```
+-----------------------+
| result                |
+-----------------------+
| all inserts succeeded |
+-----------------------+
```

Explanation: The whole script executed without a single constraint error (no ERROR lines were produced).

5.2 Row count of every table

SQL query

```
SELECT 'Pincodes' AS table_name, COUNT(*) AS row_count FROM Pincodes
UNION ALL SELECT 'Customers', COUNT(*) FROM Customers
UNION ALL SELECT 'Addresses', COUNT(*) FROM Addresses
UNION ALL SELECT 'Categories', COUNT(*) FROM Categories
UNION ALL SELECT 'Products', COUNT(*) FROM Products
UNION ALL SELECT 'Shopping_Cart', COUNT(*) FROM Shopping_Cart
UNION ALL SELECT 'Cart_Items', COUNT(*) FROM Cart_Items
UNION ALL SELECT 'Orders', COUNT(*) FROM Orders
UNION ALL SELECT 'Order_Items', COUNT(*) FROM Order_Items
UNION ALL SELECT 'Payments', COUNT(*) FROM Payments
UNION ALL SELECT 'Shipments', COUNT(*) FROM Shipments
UNION ALL SELECT 'Reviews', COUNT(*) FROM Reviews;
```

Output

```
+---------------+-----------+
| table_name    | row_count |
+---------------+-----------+
| Pincodes      |        15 |
| Customers     |        16 |
| Addresses     |        22 |
| Categories    |         6 |
| Products      |        25 |
| Shopping_Cart |        16 |
| Cart_Items    |        24 |
| Orders        |        28 |
| Order_Items   |        65 |
| Payments      |        28 |
| Shipments     |        21 |
| Reviews       |        25 |
+---------------+-----------+
```

Explanation: All twelve tables contain data.

5.3 Sample of the stored data

Customers

SQL query

```
SELECT customer_id, first_name, last_name, email, phone, registration_date FROM Customers;
```

Output

```
+-------------+------------+-----------+---------------------------+------------+-------------------+
| customer_id | first_name | last_name | email                     | phone      | registration_date |
+-------------+------------+-----------+---------------------------+------------+-------------------+
|           1 | Aarav      | Sharma    | aarav.sharma@gmail.com    | 9977446522 | 2025-01-14        |
|           2 | Priya      | Nair      | priya.nair@yahoo.in       | 6323775684 | 2025-02-03        |
|           3 | Rohan      | Mehta     | rohan.mehta@gmail.com     | 7093414184 | 2025-02-21        |
|           4 | Ananya     | Iyer      | ananya.iyer@yahoo.in      | 8886848884 | 2025-03-09        |
|           5 | Vikram     | Singh     | vikram.singh@yahoo.in     | 6375888073 | 2025-04-17        |
|           6 | Sneha      | Reddy     | sneha.reddy@yahoo.in      | 8842203407 | 2025-05-02        |
|           7 | Karthik    | Rao       | karthik.rao@gmail.com     | 7020834547 | 2025-05-28        |
|           8 | Meera      | Joshi     | meera.joshi@gmail.com     | 9923160265 | 2025-06-11        |
|           9 | Arjun      | Patel     | arjun.patel@outlook.com   | 9875805171 | 2025-07-06        |
|          10 | Divya      | Menon     | divya.menon@yahoo.in      | 9938215876 | 2025-07-30        |
|          11 | Imran      | Khan      | imran.khan@outlook.com    | 9956661846 | 2025-08-19        |
|          12 | Pooja      | Bansal    | pooja.bansal@outlook.com  | 8878932553 | 2025-09-08        |
|          13 | Siddharth  | Das       | siddharth.das@yahoo.in    | 9855686986 | 2025-09-25        |
|          14 | Neha       | Kulkarni  | neha.kulkarni@gmail.com   | 7082217824 | 2025-10-13        |
|          15 | Rahul      | Verma     | rahul.verma@outlook.com   | 9971397604 | 2025-11-04        |
|          16 | Lakshmi    | Narayanan | lakshmi.narayanan@gmail.com | 6387500451 | 2025-12-01        |
+-------------+------------+-----------+---------------------------+------------+-------------------+
```

Addresses (joined to Pincodes for readability)

SQL query

```
SELECT a.address_id, a.customer_id, LEFT(a.address_line,34) AS address_line, a.pincode, p.city, a.address_type
FROM Addresses a JOIN Pincodes p ON p.pincode = a.pincode ORDER BY a.address_id LIMIT 10;
```

Output

```
+------------+-------------+------------------------------------+---------+-----------+--------------+
| address_id | customer_id | address_line                       | pincode | city      | address_type |
+------------+-------------+------------------------------------+---------+-----------+--------------+
|          1 |           1 | 12, 4th Cross, Indiranagar         | 560034  | Bengaluru | Home         |
|          2 |           1 | Tower B, Embassy Tech Village, Out | 560066  | Bengaluru | Work         |
|          3 |           2 | 45, Residency Road, Near Brigade T | 560001  | Bengaluru | Home         |
|          4 |           3 | Flat 302, Sea View Apartments, Col | 400001  | Mumbai    | Home         |
|          5 |           3 | Nirmal Tower, Nariman Point        | 400050  | Mumbai    | Work         |
|          6 |           4 | 7, Anna Salai, Near Spencer Plaza  | 600001  | Chennai   | Home         |
|          7 |           5 | B-14, Connaught Place              | 110001  | New Delhi | Home         |
|          8 |           5 | 22, Hauz Khas Village              | 110016  | New Delhi | Other        |
|          9 |           6 | Plot 56, Banjara Hills Road No. 3  | 500001  | Hyderabad | Home         |
|         10 |           7 | 88, MG Road, Near Trinity Metro    | 560001  | Bengaluru | Home         |
+------------+-------------+------------------------------------+---------+-----------+--------------+
```

Categories

SQL query

```
SELECT * FROM Categories;
```

Output

```
+-------------+------------------+-------------------------------------------+
| category_id | category_name    | description                               |
+-------------+------------------+-------------------------------------------+
|           1 | Electronics      | Mobiles, audio, wearables and accessories |
|           2 | Clothing         | Men's and women's apparel                 |
|           3 | Footwear         | Sports, casual and formal footwear        |
|           4 | Home & Kitchen   | Appliances, cookware and storage          |
|           5 | Books            | Academic, self-help and general reading   |
|           6 | Sports & Fitness | Equipment and fitness accessories         |
```

```
+-------------+------------------+-----------------------------------------+
```

Products

SQL query

```
SELECT product_id AS id, category_id AS cat, LEFT(product_name,34) AS product_name, price, stock_quantity AS stock, brand FROM
Products;
```

Output

```
+----+-----+------------------------------------+----------+-------+----------------------+
| id | cat | product_name                       | price    | stock | brand                |
+----+-----+------------------------------------+----------+-------+----------------------+
|  1 |   1 | Redmi Note 13 5G (8GB/128GB)       | 17999.00 |    45 | Xiaomi               |
|  2 |   1 | Rockerz 450 Bluetooth Headphones   |  1499.00 |   120 | boAt                 |
|  3 |   1 | 25W Fast Charger                   |  1199.00 |   200 | Samsung              |
|  4 |   1 | M331 Silent Plus Wireless Mouse    |   899.00 |    85 | Logitech             |
|  5 |   1 | ColorFit Pulse 3 Smartwatch        |  1999.00 |    70 | Noise                |
|  6 |   2 | 511 Slim Fit Jeans                 |  2799.00 |    60 | Levi's               |
|  7 |   2 | Men's Formal Shirt                 |  1499.00 |    90 | Allen Solly          |
|  8 |   2 | Round Neck Cotton T-Shirt          |   799.00 |   150 | Puma                 |
|  9 |   2 | Women's Cotton Kurta               |  1899.00 |    55 | Fabindia             |
| 10 |   3 | Revolution 7 Running Shoes         |  3995.00 |    40 | Nike                 |
| 11 |   3 | Leather Formal Shoes               |  2299.00 |    35 | Bata                 |
| 12 |   3 | Adilette Comfort Slides            |  1799.00 |    75 | Adidas               |
| 13 |   3 | Casual Canvas Sneakers             |  1499.00 |    95 | Campus               |
| 14 |   4 | Induction Cooktop 2000W            |  2499.00 |    50 | Prestige             |
| 15 |   4 | Thermosteel Flask 1L               |   799.00 |   110 | Milton               |
| 16 |   4 | Glass Tumbler Set (6 pcs)          |   699.00 |    80 | Borosil              |
| 17 |   4 | Digital Air Fryer 4.1L             |  7999.00 |    30 | Philips              |
| 18 |   5 | Atomic Habits                      |   499.00 |   140 | Penguin Random House |
| 19 |   5 | The Psychology of Money            |   349.00 |   160 | Jaico                |
| 20 |   5 | Database System Concepts           |   899.00 |    60 | McGraw Hill          |
| 21 |   5 | Wings of Fire                      |   299.00 |   180 | Universities Press   |
| 22 |   6 | Mavis 350 Nylon Shuttlecocks (Pack |   649.00 |   100 | Yonex                |
| 23 |   6 | Football Size 5                    |   599.00 |    65 | Cosco                |
| 24 |   6 | Yoga Mat 6mm                       |   549.00 |   130 | Boldfit              |
| 25 |   6 | PVC Dumbbell Set 10 kg             |  1299.00 |    45 | Nivia                |
+----+-----+------------------------------------+----------+-------+----------------------+
```

Orders

SQL query

```
SELECT * FROM Orders LIMIT 10;
```

Output

```
+----------+-------------+------------+---------------------+--------------+--------------+
| order_id | customer_id | address_id | order_date          | order_status | total_amount |
+----------+-------------+------------+---------------------+--------------+--------------+
|        1 |          10 |         14 | 2026-01-17 10:45:00 | Delivered    |     24379.00 |
|        2 |          10 |         14 | 2026-02-20 18:56:00 | Delivered    |      4591.00 |
|        3 |          10 |         14 | 2026-03-05 20:06:00 | Delivered    |      2590.00 |
|        4 |           3 |          5 | 2026-03-12 21:39:00 | Delivered    |      3238.00 |
|        5 |          12 |         17 | 2026-04-29 16:30:00 | Cancelled    |      3768.00 |
|        6 |           1 |          2 | 2026-05-17 12:30:00 | Delivered    |     13844.00 |
|        7 |          13 |         18 | 2026-06-12 13:54:00 | Delivered    |      1499.00 |
|        8 |           3 |          5 | 2026-07-10 11:40:00 | Delivered    |      6991.00 |
|        9 |           2 |          3 | 2026-07-12 21:29:00 | Delivered    |     19439.00 |
|       10 |          14 |         20 | 2026-07-27 20:30:00 | Delivered    |     38878.00 |
+----------+-------------+------------+---------------------+--------------+--------------+
```

Order_Items

SQL query

```
SELECT * FROM Order_Items LIMIT 10;
```

Output

```
+---------------+----------+------------+----------+------------+
| order_item_id | order_id | product_id | quantity | unit_price |
+---------------+----------+------------+----------+------------+
|             1 |        1 |         13 |        2 |    1499.00 |
|             2 |        1 |          1 |        1 |   19439.00 |
|             3 |        1 |          4 |        2 |     971.00 |
|             4 |        2 |         23 |        3 |     599.00 |
|             5 |        2 |          3 |        1 |    1295.00 |
|             6 |        2 |          7 |        1 |    1499.00 |
|             7 |        3 |          3 |        2 |    1295.00 |
|             8 |        4 |          2 |        2 |    1619.00 |
|             9 |        5 |         18 |        2 |     499.00 |
|            10 |        5 |          4 |        1 |     971.00 |
+---------------+----------+------------+----------+------------+
```

Payments

SQL query

```
SELECT * FROM Payments LIMIT 10;
```

Output

```
+------------+----------+------------------+----------------+---------------------+-------------+
| payment_id | order_id | payment_method   | payment_status | payment_date        | amount_paid |
+------------+----------+------------------+----------------+---------------------+-------------+
|          1 |        1 | Card             | Successful     | 2026-01-17 10:46:00 |    24379.00 |
|          2 |        2 | Net Banking      | Successful     | 2026-02-20 18:58:00 |     4591.00 |
|          3 |        3 | UPI              | Successful     | 2026-03-05 20:12:00 |     2590.00 |
|          4 |        4 | Card             | Successful     | 2026-03-12 21:42:00 |     3238.00 |
|          5 |        5 | Net Banking      | Failed         | 2026-04-29 16:32:00 |        0.00 |
|          6 |        6 | Net Banking      | Successful     | 2026-05-17 12:33:00 |    13844.00 |
|          7 |        7 | Cash on Delivery | Successful     | 2026-06-16 18:30:00 |     1499.00 |
|          8 |        8 | UPI              | Successful     | 2026-07-10 11:46:00 |     6991.00 |
|          9 |        9 | Cash on Delivery | Successful     | 2026-07-18 18:30:00 |    19439.00 |
|         10 |       10 | UPI              | Successful     | 2026-07-27 20:31:00 |    38878.00 |
+------------+----------+------------------+----------------+---------------------+-------------+
```

Shipments

SQL query

```
SELECT * FROM Shipments LIMIT 8;
```

Output

```
+-------------+----------+--------------+-----------------+--------------+------------------------+-----------------+
| shipment_id | order_id | courier_name | tracking_number | shipped_date | expected_delivery_date | shipment_status |
+-------------+----------+--------------+-----------------+--------------+------------------------+-----------------+
|           1 |        1 | XpressBees   | XB3408887563    | 2026-01-18   | 2026-01-21             | Delivered       |
|           2 |        2 | Blue Dart    | BD1508021414    | 2026-02-22   | 2026-02-25             | Delivered       |
|           3 |        3 | XpressBees   | XB2298115953    | 2026-03-07   | 2026-03-12             | Delivered       |
|           4 |        4 | Ecom Express | ECM2619985073   | 2026-03-13   | 2026-03-17             | Delivered       |
|           5 |        6 | XpressBees   | XB7678236685    | 2026-05-18   | 2026-05-23             | Delivered       |
|           6 |        7 | Blue Dart    | BD7351583601    | 2026-06-13   | 2026-06-16             | Delivered       |
|           7 |        8 | Blue Dart    | BD9753181212    | 2026-07-12   | 2026-07-17             | Delivered       |
|           8 |        9 | Ecom Express | ECM8486523758   | 2026-07-13   | 2026-07-18             | Delivered       |
+-------------+----------+--------------+-----------------+--------------+------------------------+-----------------+
```

Reviews

SQL query

```
SELECT review_id, product_id, customer_id, rating, LEFT(comment,44) AS comment, review_date FROM Reviews LIMIT 8;
```

Output

```
+-----------+------------+-------------+--------+----------------------------------------------+-------------+
| review_id | product_id | customer_id | rating | comment                                      | review_date |
+-----------+------------+-------------+--------+----------------------------------------------+-------------+
|         1 |         13 |          10 |      5 | Excellent quality, totally worth the price.  | 2026-01-29  |
|         2 |          1 |          10 |      5 | Superb product and quick delivery!           | 2026-01-28  |
|         3 |          4 |          10 |      4 | Solid build and good performance.            | 2026-01-24  |
|         4 |          7 |          10 |      4 | Solid build and good performance.            | 2026-03-06  |
|         5 |         17 |           1 |      5 | Superb product and quick delivery!           | 2026-05-29  |
|         6 |         24 |           1 |      2 | Quality could be better.                     | 2026-06-04  |
|         7 |          9 |           1 |      4 | Good value for money, minor packaging issue. | 2026-05-28  |
|         8 |          7 |          13 |      4 | Works well. Would recommend.                 | 2026-06-25  |
+-----------+------------+-------------+--------+----------------------------------------------+-------------+
```

Shopping_Cart and Cart_Items

SQL query

```
SELECT c.cart_id, c.customer_id, c.created_date, COUNT(ci.cart_item_id) AS items
FROM Shopping_Cart c LEFT JOIN Cart_Items ci ON ci.cart_id = c.cart_id GROUP BY c.cart_id, c.customer_id, c.created_date;
```

Output

```
+---------+-------------+--------------+-------+
| cart_id | customer_id | created_date | items |
+---------+-------------+--------------+-------+
|       1 |           1 | 2025-01-25   |     3 |
|       2 |           2 | 2025-02-19   |     3 |
|       3 |           3 | 2025-02-25   |     2 |
|       4 |           4 | 2025-04-06   |     0 |
|       5 |           5 | 2025-05-14   |     2 |
|       6 |           6 | 2025-05-26   |     2 |
|       7 |           7 | 2025-05-31   |     0 |
|       8 |           8 | 2025-06-24   |     3 |
|       9 |           9 | 2025-07-15   |     0 |
|      10 |          10 | 2025-08-19   |     2 |
|      11 |          11 | 2025-08-29   |     0 |
|      12 |          12 | 2025-09-29   |     3 |
|      13 |          13 | 2025-09-29   |     2 |
|      14 |          14 | 2025-10-25   |     2 |
|      15 |          15 | 2025-11-20   |     0 |
|      16 |          16 | 2025-12-31   |     0 |
```

```
+---------+-------------+--------------+--------+
```

5.4 Integrity checks on the loaded data

Orders whose total does not equal the sum of their lines (expected: 0)

SQL query

```
SELECT COUNT(*) AS mismatched_orders FROM (SELECT o.order_id FROM Orders o JOIN Order_Items i ON i.order_id=o.order_id
 GROUP BY o.order_id, o.total_amount HAVING o.total_amount <> SUM(i.quantity*i.unit_price)) x;
```

Output

```
+-------------------+
| mismatched_orders |
+-------------------+
|                 0 |
+-------------------+
```

Explanation: Every stored total agrees with its line items.

Orphan check – child rows without a parent (expected: all 0)

SQL query

```
SELECT (SELECT COUNT(*) FROM Orders o LEFT JOIN Customers c ON c.customer_id=o.customer_id WHERE c.customer_id IS NULL) AS
orders_without_customer,
       (SELECT COUNT(*) FROM Order_Items i LEFT JOIN Products p ON p.product_id=i.product_id WHERE p.product_id IS NULL) AS
items_without_product,
       (SELECT COUNT(*) FROM Payments y LEFT JOIN Orders o ON o.order_id=y.order_id WHERE o.order_id IS NULL) AS
payments_without_order,
       (SELECT COUNT(*) FROM Reviews r LEFT JOIN Customers c ON c.customer_id=r.customer_id WHERE c.customer_id IS NULL) AS
reviews_without_customer;
```

Output

```
+------------------------+----------------------+-----------------------+--------------------------+
| orders_without_customer | items_without_product | payments_without_order | reviews_without_customer |
+------------------------+----------------------+-----------------------+--------------------------+
|                      0 |                    0 |                     0 |                        0 |
+------------------------+----------------------+-----------------------+--------------------------+
```

Explanation: The foreign keys guarantee there are no orphans; the query confirms it.

Business-rule check – every order uses an address that belongs to the ordering customer (expected: 0 violations)

SQL query

```
SELECT COUNT(*) AS violations FROM Orders o JOIN Addresses a ON a.address_id = o.address_id WHERE a.customer_id <> o.customer_id;
```

Output

```
+------------+
| violations |
+------------+
|          0 |
+------------+
```

Explanation: The composite foreign key makes this rule unbreakable.

Status distribution of orders, payments and shipments

SQL query

```
SELECT 'Orders' AS entity, order_status AS status, COUNT(*) AS n FROM Orders GROUP BY order_status
UNION ALL SELECT 'Payments', payment_status, COUNT(*) FROM Payments GROUP BY payment_status
UNION ALL SELECT 'Shipments', shipment_status, COUNT(*) FROM Shipments GROUP BY shipment_status;
```

Output

```
+-----------+------------+----+
| entity    | status     | n  |
+-----------+------------+----+
| Orders    | Cancelled  |  3 |
| Orders    | Delivered  | 17 |
| Orders    | Placed     |  4 |
| Orders    | Shipped    |  4 |
| Payments  | Failed     |  3 |
| Payments  | Pending    |  2 |
| Payments  | Successful | 23 |
| Shipments | Delayed    |  1 |
| Shipments | Delivered  | 17 |
| Shipments | In Transit |  3 |
+-----------+------------+----+
```

Explanation: The data set contains every status value allowed by the CHECK constraints, which makes the later reports meaningful.

6. Conclusion

All twelve tables were populated with 270+ interrelated rows in a single run with zero constraint violations. The verification queries confirm correct row counts, consistent order totals, no orphan records and a balanced mixture of statuses, so the database is ready for the SQL queries and reports in Assignments 8 and 9.